

YANHAN RUAN

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🎓 EDUCATION

Wuhan University (WHU), Wuhan, China Sept.2022 – Dec.2024

Master of Science degree in Digital Water Systems and Intelligent Solutions

Northwest A&F University (NWAUFU), Shaanxi, China Sept.2018 – Jul.2022

Bachelor of Science degree in Water Resources and Hydropower Engineering

👥 EXPERIENCE & PROJECTS

Carbon Neutralization Big Data Platform, Shanghai, China Jun.2024 – Sept.2025

TypeScript | React | Express | Three.js | Resium FullStack Developer

- **Optimized algorithm performance** by implementing a dual-pointer approach, reducing time complexity from $O(n^2)$ to $O(n)$ and accelerating remote sensing image preprocessing by 150%.
- **Enhanced rendering efficiency** by encapsulating reusable utility hooks, throttling Cesium re-render events, and optimizing entity loading within specific viewport heights.
- **Secured application components** by developing robust Higher Order Components (HOC) for streamlined login validation and authentication routing.

e-idol (Full-Stack Gaming Marketplace), Tokyo, Japan Oct.2025 – Now

React | Tailwind CSS | Go (Gin) | PostgreSQL | WebSockets | GORM Personal Project

- **Architected a full-stack marketplace** using Go and React, securing client-host interactions with JWT and Role-Based Access Control (RBAC).
- **Engineered a concurrent booking system** utilizing database-level transactional locks (FOR UPDATE), successfully eliminating race conditions and double-booking.
- **Built real-time communication** via native Go WebSockets (Hub/Client model) for instant messaging and seamless order state updates.
- **Optimized frontend rendering** by developing a custom HTML5 Canvas engine with requestAnimationFrame, replacing heavy third-party animation libraries.
- Implemented a context-based **i18n** system enabling multilingual localization.

shadcn/ui | Component Contributor Feb.2026

TypeScript | React | Tailwind CSS | Playwright Open Source

- **Resolved a critical Safari rendering collapse** in the Calendar component by refactoring the hybrid layout architecture from Flexbox to CSS Grid, ensuring pixel-perfect cross-browser consistency.
- **Engineered a cross-platform debugging workflow** utilizing Playwright's WebKit engine on Windows to isolate macOS-specific CSS anomalies, culminating in a successful Pull Request to the repository.

📁 SKILLS & AWARDS

- Languages & Frameworks: TypeScript, ECMAScript 6+, Go, Python, React, Next.js, Node.js, Gin
- Cloud & Specialized Technologies: Tailwind CSS, shadcn/ui, Webgis, NeRF, AWS, Docker, CI/CD
- 2022 ZJU PAT - Professional Ability Test (95 / 100 score) • 2022 ACM-ICPC excellence-award
- 2020 MCM S-Award • 2020 CMC second-prize

🌐 LANGUAGES

- English: TOEIC 805; Professional proficiency for workplace communication and technical documentation.
- Japanese: JLPT N2 certified; conversational proficiency for daily communication.